

Commissioning of LUNA three-dimensional surface-guided radiotherapy in full rotating gantry proton therapy system

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ABSTRACT

Background: Surface-guided radiotherapy (SGRT) is a real-time, non-ionizing optical imaging technology that monitors patient surfaces during radiation treatment by creating 3D reconstructions from surface coordinate measurements. LAP has recently released a new SGRT solution, and we are the first proton therapy center to adopt this SGRT solution in a full rotating gantry. This work aims to present the commissioning results and quality assurance procedures at our center and to highlight the challenges of installing SGRT on a full rotating gantry.

Methods: Following closely the recommendations in TG-302, 6 different measurements were performed during the commissioning of the SGRT system, which was mounted linearly on the same beam due to space constraints. These measurements included spatial drift measurement, static localization accuracy, dynamic localization accuracy, end-to-end localization accuracy, field-of-view (FOV) characterization of different treatment sites, and stability upon gantry rotation. These measurements subsequently served as baseline values for the different quality assurance procedures recommended in TG-302.

Results: The moving average for spatial drift measurement across 200 minutes was 0.0233 mm, while the average deviation for static and dynamic localization was less than 0.2 mm across. The largest deviation was 0.182 mm in the LR direction at 40 mm for static localization, while dynamic localization has a maximum deviation from the Anzai laser tracking at 0.1 minute point with a value of 0.734 mm. End-to-end localization test reported a difference of 0.08 cm between x-ray imaging and LUNA 3D's isocenter. The FOV was in general mostly visible for all treatment sites except for the lower neck region and lateral prostate surfaces due to the placement of the cameras. Stability during gantry rotation is kept within 0.2 mm with gantry angle 30.4° and 335.8° having the largest magnitude deviations of 0.151 and 0.156 mm, respectively.

Conclusion: The successful commissioning of the LUNA 3D SGRT system at our center can serve as a reference for other particle therapy centers with a full rotating gantry. This work will allow them to include the relevant tests as well as understand the limitations of FOV for different treatment sites when adopting similar camera arrangement like ours in a full rotating gantry.

Introduction

Surface-guided radiotherapy (SGRT) utilizes optical surface imaging techniques for real-time, non-ionizing, and non-invasive monitoring

and evaluation of a patient's surface during radiotherapy treatment. The overarching objective is to accurately measure the coordinates of points on a patient's external body, translate this data into a depth map, and generate a three-dimensional (3D) reconstruction. This is followed

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by registration of the generated surface with the reference surface using a proprietary surface registration algorithm. In response to the growing interest in SGRT, several review articles^{1–4} have been published, alongside the 2 main recommendations for the clinical implementation of SGRT systems: the AAPM TG-302 report⁵ and the ESTRO-ACROP guidelines.⁶

The main advantages of SGRT include its non-ionizing nature and improvements in the speed and accuracy of patient setup,^{7,8} as well as real-time intra-fractional motion monitoring during beam delivery. These are achieved through tattoo-free workflows^{9,10} and are attributed to the system's large field-of-view (FOV) compared to other imaging modalities, such as Hiatchi's large FOV cone beam computed tomography, which has a FOV of $48\text{cm} \times 25\text{cm}$ (LatxLong). Some SGRT systems are also equipped with breath-hold or gating capabilities for the treatment of tumors under respiratory motion.^{11,12}

Within the context of proton therapy, SGRT has shown promise as an effective pre-positioning tool prior to CT or cone beam computed tomography imaging, as well as for its potential to reduce overall treatment time by optimizing the frequency of repeat imaging. These advantages are particularly salient in pediatric radiotherapy, where minimizing cumulative imaging dose is paramount,^{13–16} especially given that proton therapy is generally regarded as the preferred treatment modality for pediatric patients.^{17,18} Nonetheless, the integration of SGRT into proton therapy presents a unique set of challenges. Among these is the potential occlusion of SGRT cameras by the nozzle in full rotating proton gantry systems, which may limit the FOV and, in turn, compromise the fidelity of surface tracking and acquisition.

As the clinical implementation of SGRT in proton therapy remains in its early stages, the literature on its commissioning practices is still limited. In this work, we delineate the commissioning and clinical implementation of the first LUNA 3D SGRT system (LAP GmbH Laser Applikationen, Germany) and its integration within a full rotating proton therapy gantry. While comprehensive commissioning results on various SGRT camera configurations, such as the 3 camera configurations for conventional C-arm linac, 5

camera configuration for half-gantry room as well as in-bore SGRT for closed-bore linacs can be found online, information on the use of a linear camera arrangement with a full rotating gantry and limited mounting space available due to the gantry drum is not available.^{19,20}

Materials and methods

Introduction to LUNA 3D hardware and software

The LUNA 3D adopts stereovision-based optical imaging technology.^{4,21,22} Stereovision-based imaging systems operate based on stereo-photogrammetry, which utilizes 2 or more camera units, each consisting of a projector and image sensor, to triangulate and extract depth information following the projection and reflection of a pseudo-random speckled light pattern on a patient's surface. The pseudo-random projection is shown in Figure 1B explicitly. The wavelength of the light pattern used by LUNA 3D is 465 nm, and to minimize any discomfort to patients and ensure that the light does not impair the therapist's work in setting up the patient accurately to the orthogonal lasers in the room, the light is switched off during initial patient set-up. Currently, there is no negative feedback from patients or therapists regarding discomfort from the light. LUNA 3D has a field-of-view (FOV) of $700 \times 1300 \times 600\text{mm}^3$ (Lat $\times$ Long $\times$ Vert) in the clinical setting and can be further increase beyond $1500 \times 1500 \times 1500\text{mm}^3$ (Lat $\times$ Long $\times$ Vert) in the research setting. The frame rate is dependent on the region of interest (ROI) size but is typically greater than 12 Hz. LUNA 3D employs rigid registration for comparison between the reference and measured surface to determine the translational and rotational differences. This is similar to the registration algorithm used by VisionRT. On the other hand, C-RAD offers a non-rigid registration algorithm with the aim of obtaining a better match for surfaces that are around the isocenter.²³ All the processing is performed using graphical processing units.

A screenshot of the software is shown in Figure 2. The user's interaction with the SGRT software operation occurs during 1) pre-treatment preparation and 2) treatment monitoring. During pre-

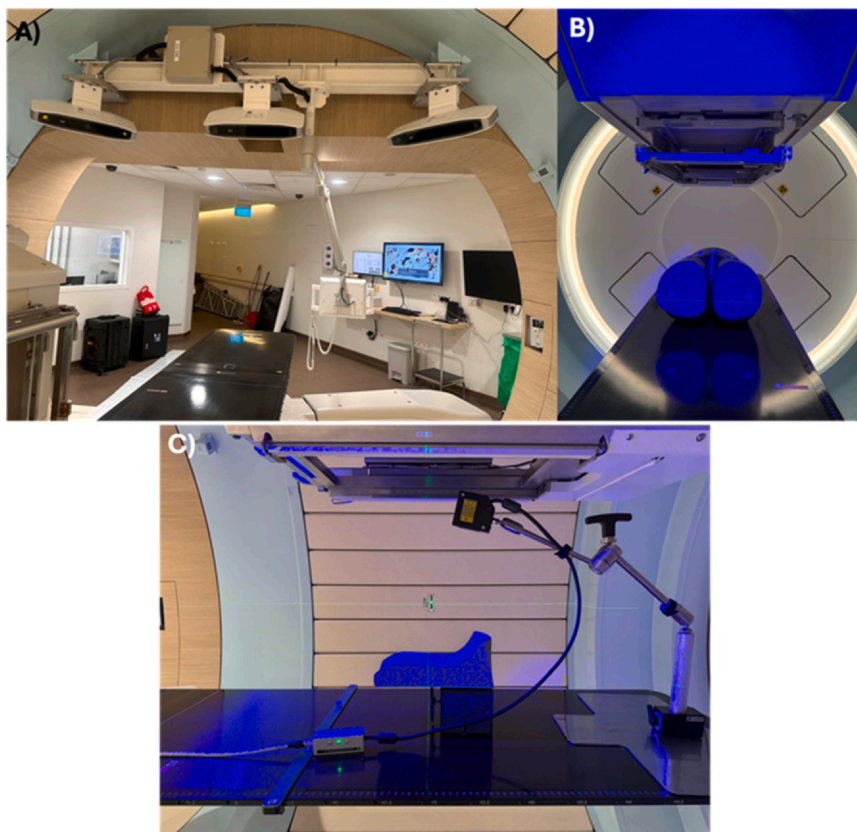


Figure 1. Photos of the LUNA 3D hardware setup and experimental designs. (a) LUNA 3D camera setup in the proton gantry. (b) Experimental setup using a pelvis phantom for localization accuracy measurement. (c) Experimental setup for dynamic localization measurement using the commissioned Anzai laser measurement as the ground truth.

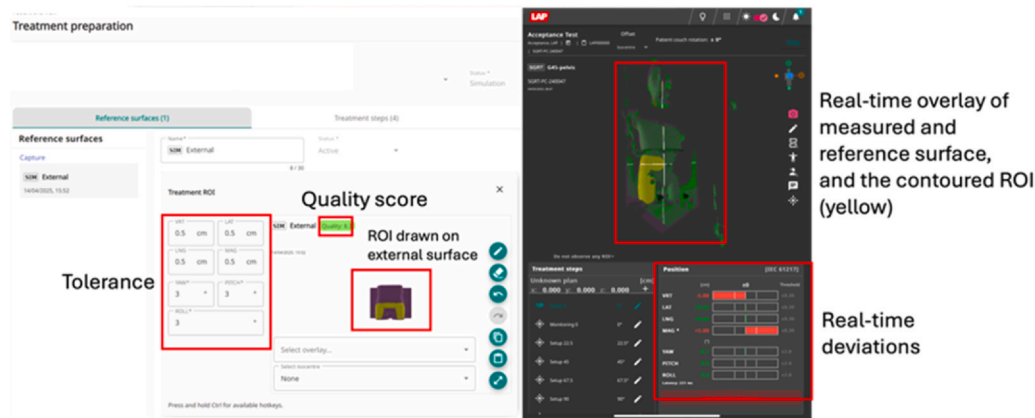


Figure 2. Photos of the LUNA 3D software interface. Software interface of the LUNA 3D SGRT with the left panel showing the pre-treatment interface while the right panel shows the treatment monitoring interface.

treatment preparation, the DICOM plan and structure set are imported into the LUNA 3D database. The reference surface is assigned to the external body created during treatment planning in the DICOM structure set. It is important to exclude the immobilization device from the external body during contouring to ensure the reference surface is anatomically relevant. The ROI is also drawn on the reference surface during pre-treatment preparation. The ROI is important as the translation and rotational shifts are calculated based on the ROI between the reference and measured surfaces. There is a “quality score” assigned to the ROI to indicate how well the system is able to track the ROI. It is strictly based on the degree of topographical features within the ROI, with a maximum score of 10.0 representing a high degree of features for reliable shift calculations. During treatment, the reference surface and the ROI defined during pre-treatment are displayed and compared with the real-time measured surface to determine the shifts. Apart from displaying the numerical values of the shifts, LUNA 3D has developed the concept of a “virtual laser.” The virtual lasers are shown as red and blue lines in the software for the measured and reference surfaces, respectively. The radiation therapists can use these lines as a guide to align the patient, which is particularly useful for correcting rotational errors. When the virtual lasers are aligned, a white line is shown to indicate patient alignment. Lastly, it is also possible to set the daily measured surface as the reference surface, and this feature is important if the external anatomy of the patient has changed drastically from simulation time.

Introduction to the proton therapy facility and LUNA 3D installation

The LUNA 3D SGRT was installed in one of the proton therapy gantries in the National Cancer Centre Singapore in March 2025. National Cancer Centre Singapore uses the Hitachi Probeat, which is synchrotron-based proton therapy system with a full rotating gantry.

The ideal LUNA 3D SGRT system setup comprises one camera located at the inferior end of the couch and 2 cameras being located laterally on both sides of the couch. However, due to the constraints of the full rotating gantry, it was challenging to place the 2 cameras at the lateral positions, so an alternative placement was adopted as shown in Figure 1A. All three cameras are mounted linearly on a beam at a height and longitudinal distances of 1.2 and 1.3 m, respectively, from the isocenter. Each camera weighs approximately 5 kg, and the beam was structurally stable to support the cameras.

SGRT commissioning

The SGRT commissioning focused on measurements that were pertinent for the deployment of SGRT in clinical operation involving patient setup and intra-fraction monitoring. There was no gating interface

with the proton therapy at the point of commissioning, which removed the commissioning aspects involving gated treatment. The commissioning items were decided based on the TG-302 and ESTRO-ACROP guidelines. The measurements included spatial drift measurement, static and dynamic localization accuracy, FOVs, and stability upon gantry rotation. LUNA 3D has the option to export a log file containing the time series of all real-time position measurements captured with a sampling rate of greater than 12 Hz. In-depth analyses of the translational and rotational shifts were carried out using the time-resolved log file data.

Spatial drift measurement: The pelvis phantom was first set up as shown in Figure 1B, and a ROI was drawn on it such that it achieved a quality score of 10.0. The SGRT cameras were turned off the day before and measurements were performed immediately after turning them on. The measurements were taken continuously for three hours to assess spatial drift in the reported translational errors.

Static localization accuracy: Using a similar setup as shown in Figure 1B, the couch was translated in all three directions up to $\pm 10.0\text{cm}$, the pitch and roll were varied up to $\pm 5^\circ$ and the yaw was varied up to $\pm 90^\circ$. The localization accuracy was quantified by the difference between the measured positions or rotations and the preset couch positions. The translational and rotational accuracy of the couch was validated according to TG-224.

Dynamic localization accuracy: This measurement was achieved using a Varian motion phantom that provides a sinusoidal motion and a headrest with topographical features as shown in Figure 1C. The Anzai respiratory gating system, AZ-733VI (Anzai, Tokyo, Japan), which had been commissioned previously and was in clinical operation, was set up to provide the ground truth measurement using the laser sensor as shown in the same figure. The laser was positioned parallel to the motion to ensure that the measured motion by Anzai corresponded to the vertical motion of the SGRT. The smallest possible ROI (while maintaining perfect score) was drawn on the surface to represent the same location as the Anzai laser spot. The stage was set in motion, and both the SGRT and Anzai recorded the time series of the positions for 1 minute. After this, the log files from both systems were extracted and compared to determine the dynamic localization accuracy in the AP direction.

End-to-end localization accuracy: The same pelvis phantom attached with markers was scanned, and a treatment plan with the isocenter set at the markers was generated and imported into the SGRT system. Using x-ray imaging acquired orthogonally at 0° and 90° , the pelvic phantom was set up, and the position was corrected based on the markers. The process of setting up and aligning the phantom was repeated and done by three different people. The values shown by the SGRT system were then taken as the difference between the isocenters of the x-ray imaging and the SGRT system.

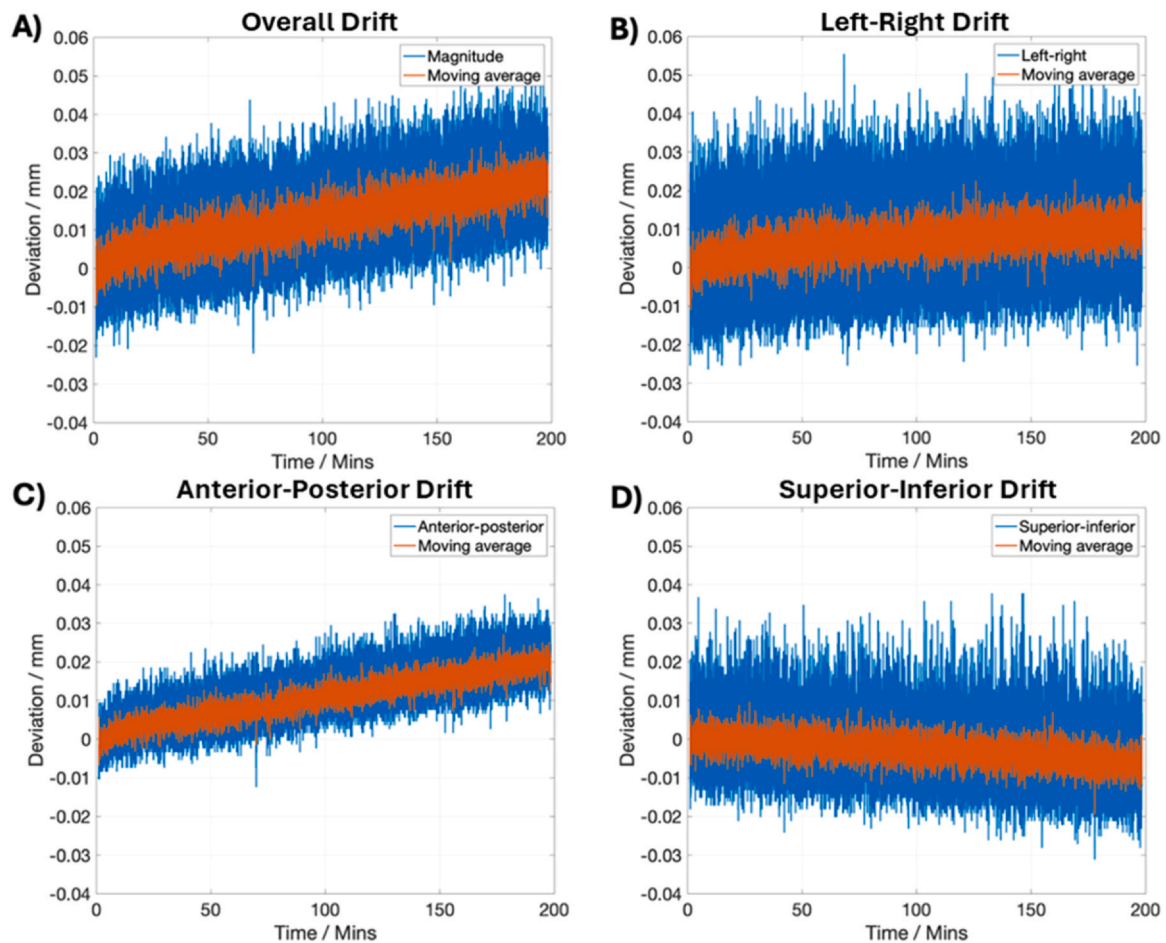


Figure 3. Spatial drift measurements, Shows the overall spatial drift measurement results over three hours, with the raw data in blue and the 20 samples moving average in orange. (A) Overall drift, (B) left-right drift, (C) anterior-posterior drift, and (D) superior-inferior drift.

Stability upon gantry rotation: This measurement used the same setup as Figure 1B. The gantry was rotated through 360° , and the translational and rotational deviations were recorded during the rotation.

Field-of-view: This measurement aims to shed light on the FOV achievable for different treatment sites, namely, head-and-neck, breast, thorax, and pelvis. An ATOM model 705 anthropomorphic phantom (Sun Nuclear Corporation, USA) was used for this measurement. The breast inserts were only used for the breast treatment site, while they were removed for all the remaining 3 sites. Realistic isocenters were also selected based on our treatment planning practice, which resulted in a different isocenters positions in all 4 sites, particularly in the SI direction.

Results

Spatial drift measurement

The results are shown in Figure 3. Figure B, C and D show the drifts in the left-right (LR), anterior-posterior (AP), and superior-inferior (SI) directions. Figure A shows the Euclidean magnitude calculated from the 3 directions. The orange lines show the moving average with a window of 10 data points. The maximum drift across 200 minutes is less than 0.05 mm in magnitude and the three directions. The magnitude drifts at a rate of 0.0233 mm in 200 minutes (from the orange line). The drift observed in the AP direction contributed the most to the spatial translational drift, while the drift in the SI direction contributed the least to it.

Static localization accuracy

The results for rotational accuracy are shown in Figure 4A and B. Figure 4B shows the range from -5° to 5° while Figure 4A shows up to $\pm 90^\circ$ for the yaw when treating at non-coplanar angles. The accuracies are within 0.1° with the largest being 0.0578° at a 75° yaw. The roll and pitch accuracies are within 0.05° across all the tested angles. Figure 4C shows the translational accuracy. The average deviation is less than 0.2 mm in general, with the largest mean error being 0.182 mm in the LR direction at 40 mm (blue markers).

Dynamic localization accuracy

The comparison of the real-time positions of the moving phantom measured by LUNA 3D and Anzai laser system is shown in Figure 5. Figure 5A shows the overlay of the two displacement curves, and Figure 5B shows the difference between the displacements. The relative time offset between the two curves were optimized to decrease the overall magnitudes of the displacement difference in Figure 5B. The shapes of the two displacement curves were very similar in Figure 5A with an average difference of 0.150 mm and a maximum difference of 0.734 mm at the 0.1 minute mark.

End-to-end localization accuracy

Table 1 shows the results for the end-to-end localization accuracy test. The values are all within the tolerance of 0.1 cm, with the maximum magnitude of 0.08 cm. The largest difference was found in the LR

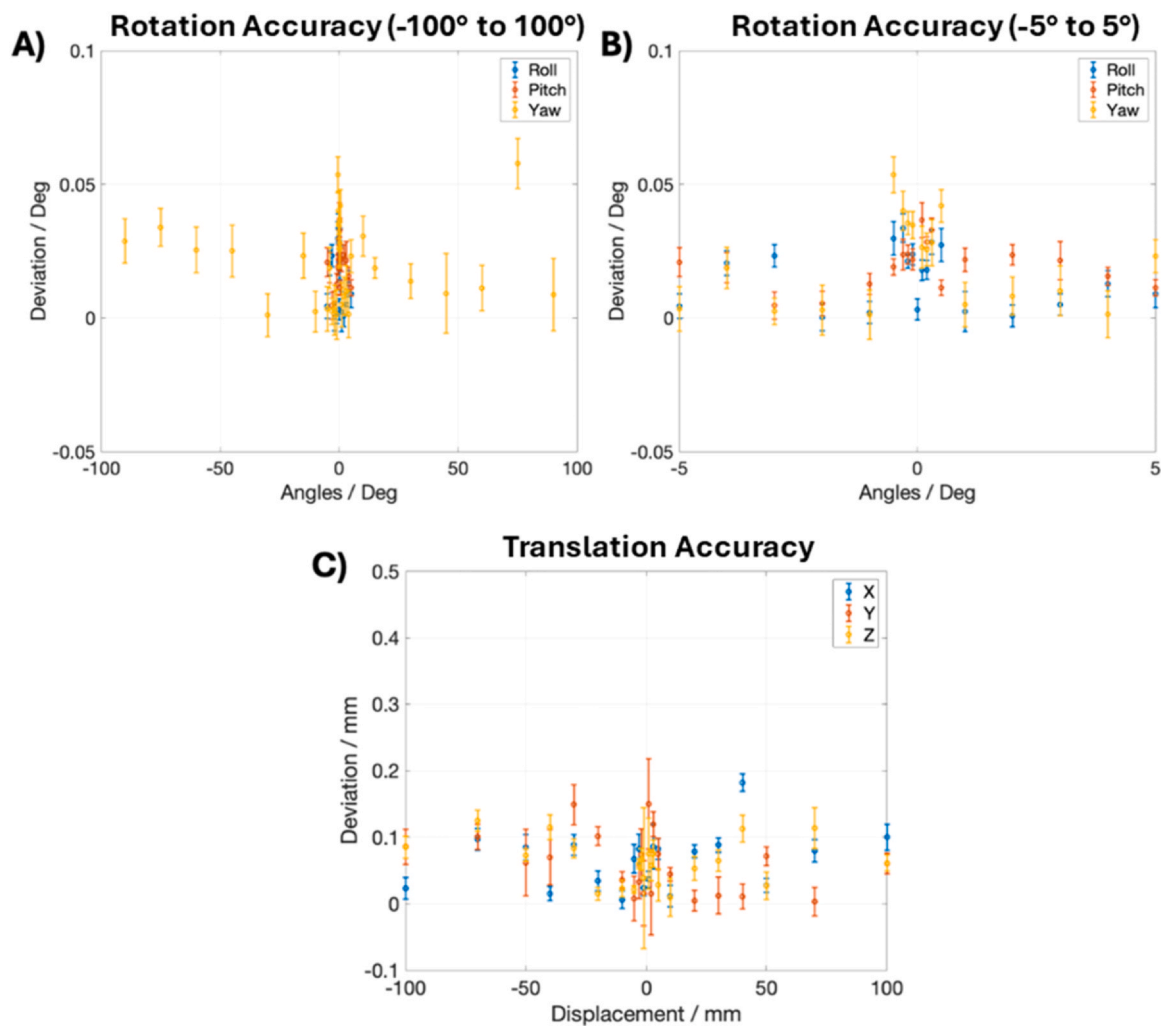


Figure 4. Static localization accuracy results. Shows the static localization accuracy results the SGRT system. (A) Static localization accuracy results for roll, pitch, and yaw at $\pm 100^\circ$ with one standard deviation as the error bar. (B) Static localization accuracy results for roll, pitch, and yaw at $\pm 5^\circ$ with one standard deviation as the error bar. (C) Static localization accuracy for translation with one standard deviation as the error bar.

direction with a magnitude of 0.08 cm, with the overall magnitude having an average value of 0.06 ± 0.02 cm.

Field-of-view characterization of different treatment sites

The FOVs for different sites are shown in Figure 6. The green regions are surface on the anthropomorphic phantom, which are visible to the SGRT system, while the black regions cannot be observed from the cameras. In the breast, thorax, and head-and-neck treatment sites, the areas superior to the isocenter and around the supraclavicular regions are black in color, which are not visible to the camera. This is due to the nature of the contour of the chest, which resulted in the shadowing of the projected light. On the other hand, the pelvis treatment site has a full FOV extending at least 20 cm in the superior and inferior directions from the isocenter without any invisible regions around the isocenter. Apart from the anterior surface, the lateral surface is also not entirely visible for all three treatment sides, especially for areas posterior to the isocenter. This is illustrated by the black region on the left side of the phantom in the pelvis treatment site but is similar for the rest of the sites.

Stability upon gantry rotation

Figure 7 shows the recorded deviations in the positions calculated by the LUNA 3D while the gantry is rotating in the LR, SI, and AP directions. The final Euclidean magnitude is also calculated and shown in

Figure 7A. The moving average over 10 data points are also shown as the orange curve. The largest magnitude deviation is 0.156 mm at gantry angle 335.8° and 0.151 mm at 30.4° . The average magnitude deviation is 0.053 mm. The major contributions to the magnitude deviation were from the LR and AP directions, as shown in Figure 7B and D. The largest deviations in the LR and AP directions also occur at about 30° and 330° .

Discussion

In this work, a total of 6 measurements were performed for commissioning, which comprised: 1) Spatial drift measurement, 2) Static localization accuracy, 3) End-to-end localization accuracy, 4) Dynamic localization accuracy, 5) Field-of-view characterization of different treatment sites, and 6) Stability upon gantry rotation.

In the first measurement, the spatial drift of the system was measured over a total of 200 minutes. The vendor-recommended stabilization period is 180 minutes when the cameras have been off for an extended period. Examining the overall moving average drift after the 180-minute stabilization period, the values remain relatively constant at less than 0.03 mm. The overall drift magnitude of less than 0.05 mm is significantly smaller than values reported by Mhatre²⁴ (0.7 mm) and Lehmann et al²⁵ (1.2 mm) for comparable systems. While the AP direction contributes more to the overall drift magnitude, the drift remains relatively small overall. These results are within TG-302

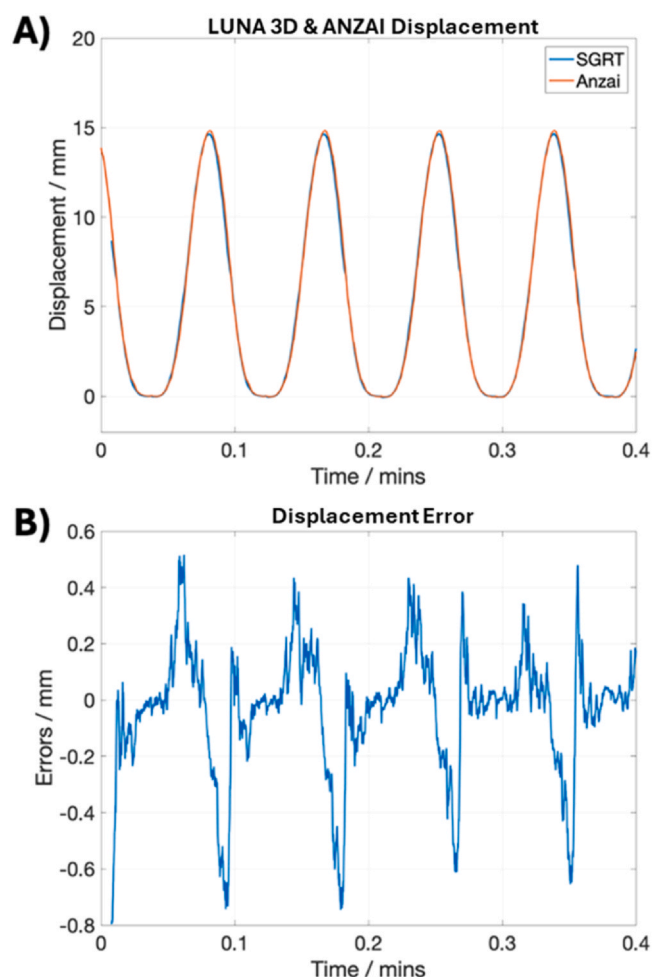


Figure 5. Dynamic localization results. (A) Raw measurement from the log files obtained from LUNA 3D (in blue) and ANZAI (in orange), respectively. (B) Errors between LUNA 3D and ANZAI.

Table 1

End-to-end localization accuracy results from 3 independent set up.

	1	2	3
Left-Right/cm	−0.02	−0.08	−0.03
Anterior-Posterior/cm	0.04	0.02	0.00
Superior-Inferior/cm	−0.01	−0.01	−0.03
Overall Magnitude/mm	0.05	0.08	0.04

recommendations, with drift values within the recommended tolerance of ≤ 2 mm over 1 hour and ≤ 1 mm after the stabilization period.

For static localization accuracy, we followed the tolerance of ≤ 1 mm, and our measured deviation of 0.2 mm and 0.1° falls well within this tolerance. Across the rotational results for pitch, roll, and yaw, roll demonstrated the highest accuracy, while yaw showed the least accuracy. During pitch and roll motions, surface displacement is more readily detected by the cameras compared to yaw motion, which appears as movement across the camera field of view rather than toward or away from the cameras. For translational testing, deviations remained within 0.2 mm with no clear directional trend across the three axes. The tested displacement range of ± 10 cm is typically sufficient to cover the entire clinical treatment region positioning requirements.

The SGRT system is also compared against a clinical system currently in use at our center. The ANZAI laser system is capable of amplitude-based respiratory gating, which provides us with the absolute motion measurements of the RGSC moving phantom. The laser is

positioned to target the same surface location on the head rest as the ROI defined in the SGRT system, ensuring spatial correspondence between the 2 measurement systems. Over a total displacement range of 15 mm, the maximum disagreement between the SGRT and ANZAI laser systems is less than 1.0 mm, which is clinically acceptable and well within the tolerance limits typically used for surface-guided gating applications. This level of agreement demonstrates excellent correlation between the 2 independent measurement systems and validates the dynamic localization accuracy of the SGRT system.

In the end-to-end localization test, the phantom was first set to the isocenter of the imaging system before collecting the positional difference from the SGRT system. This allowed us to compare the isocenter of the SGRT system against the x-ray imaging system. The result obtained were within the stricter tolerance of ≤ 0.1 cm found in TG-302. The maximum magnitude obtained is similar to those reported by Otes et al. when commissioning a 5-camera SGRT system in a proton half gantry room, where they recorded a maximum absolute difference of 0.08 cm as well.¹⁹

In this commissioning test, the FOV for various treatment sites was also tested using an anthropomorphic phantom. Due to the camera mounting position, some surface information is missing from the different sites tested. For the head and neck treatment region, the lower neck and chest areas are undetected due to the natural curvature of the human body. In the case of head and neck treatments, the primary treatment region, which can range from the nasal cavity to the larynx, remains visible to the cameras. Only the lymph nodes in the lower neck treatment region are not visible and would require either the use of complementary imaging such as X-ray verification or a more robust immobilization protocol around the shoulder region to prevent motion.

The thorax treatment site suffers from the same occlusion, and as such, the use of the SGRT system is limited depending on the target position. Ideally, tumors which are posterior from around the heart onwards will benefit from the use of the SGRT system for motion management or intrafraction monitoring.

In breast treatment regions, we can observe that the presence of intact breasts provides topographical features that can be readily detected by the SGRT system, resulting in more surface of the chest being detected. The natural curvature and prominent surface variations of intact breast tissue create topographical features that enable the SGRT system to track effectively. In contrast, post-mastectomy treatments present significantly greater challenges for SGRT implementation, as the flattened chest wall lacks the distinctive topographical features necessary for reliable surface tracking. To address this limitation, one effective strategy implemented at our center is the expansion of the ROI to include anatomical structures with more prominent topographical features. Specifically, we extend the ROI to encompass the arms-up position, which increases the variation in surface as the axillary regions and shoulder contours can be included for monitoring.

Moving on to the pelvis treatment region, although the lateral surfaces are not visible to the SGRT system, the large FOV of the SGRT systems is beneficial for patient setup as well as intrafraction monitoring during treatment. The use of surface guidance in the pelvis treatment region could be of limited utility as tumors in this region are normally deep-seated, and as such, the surface motion might not accurately represent the actual internal motion.^{26,27} As such, SGRT in the pelvis region plays a secondary IGRT role, and radiographic imaging serves as the primary IGRT tool.

In the stability upon gantry rotation test, the final Euclidean magnitude was calculated with the largest deviation lesser than 0.2 mm, which is similar to the values measured in the static localization accuracy test. When compared to camera occlusion by a C-arm linac, Wiant et al²⁸ report a worst case of less than 0.2 mm, while Peng et al²⁹ reported a deviation of 0.4 mm. The largest deviation peaks at two different angles, 335.8° and 30.4° , which corresponds to the position where the nozzle of the gantry is potentially blocking either one of the three cameras of the system. This results in a reduction in information

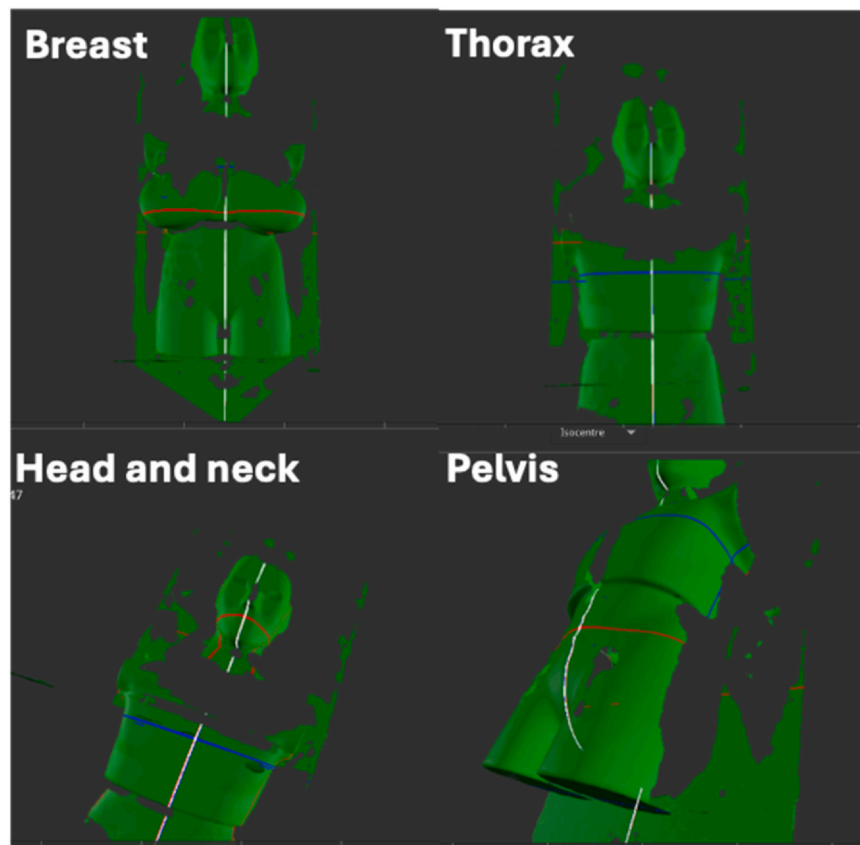


Figure 6. FOVs for various treatment sites. This figure displays the FOV for various treatment sites using an anthropomorphic phantom. The red line represents the actual isocenter positions, and the green surface represents the observable surface from the SGRT camera. The black regions are the undetected areas.

that is available for the calculation of the object's position. The deviation of position due to the gantry can be seen in all 3 directions, with the superior-anterior being the least affected. Furthermore, due to the limitations on the position in which the cameras can be installed on, they are mounted onto the I-beam, which is fixed on to the enclosure of the gantry. The camera units are thus more susceptible to vibrations during the rotation, which manifested as a larger noise in the measurements.

These five measurements establish the baseline performance characteristics required for the different proposed QA procedures. These procedures were adopted with reference to TG-302 guidelines as well as from the vendor's recommendations. The table of test and frequency can be found in Table 2.

Daily QA is performed using the vendor-provided LUNA 3D daily QA phantom for deviation between the SGRT system's isocenter and the room isocenter. This process utilizes a spotted pattern board placed on the treatment couch that is detected by all three cameras. The daily verification procedure measures deviations from the calibrated position with an internally established tolerances of ≤ 0.5 mm and $\leq 0.2^\circ$, which are more stringent than the TG-302 recommended tolerance of ≤ 2.0 mm.

Monthly QA follows TG-302 recommendations and includes both static and dynamic localization tests. The static localization test specified in TG-302 is performed by comparing couch position readouts against the positional shifts displayed in the SGRT system. This is followed by the dynamic localization test, which compares real-time phantom positions measured by the LUNA 3D system against reference measurements from the Anzai laser system to assess the system's ability to track motion accurately. The tolerance for both tests is set at 1.0 mm, which aligns with the commissioning test acceptance threshold of less than 1.0 mm.

Annual QA includes end-to-end localization test, spatial drift measurement, and stability upon gantry rotation testing. Since our current

treatment delivery uses only static gantry angles with new reference images acquired at each specific treatment angle, the gantry rotation stability test is appropriately performed on an annual basis.

Overall, the LUNA 3D SGRT system demonstrated exceptional tracking performance that significantly exceeded TG-302 recommendations, with overall deviations of less than 0.2 mm and 0.1° for both static localization and stability during gantry rotation—well within the recommended tolerance TG-302 report. This first report on the LUNA 3D system with a Hitachi Probeat full gantry installation required the addition of stability testing during gantry rotation, addressing the unique challenges of a full gantry proton environments, where gantry movements create vibrations that impact the SGRT system's performance. The linear camera arrangement at different angles, while restricting field-of-view compared to traditional lateral camera setups, successfully overcame the typical installation constraints in proton gantry that often lack suitable mounting locations. The availability of solutions available to address the FOV restrictions, combined with the superior tracking performance and workflow efficiency gains, establish this configuration as a viable solution for SGRT integration in challenging proton therapy environments with full gantry systems.

One of the limitations of this commissioning test is the absence of a full end-to-end testing. This is because the SGRT system does not provide any interfacing to the treatment machine and couch which makes it challenging to test implement a full SGRT workflow, including couch shifts to the treatment machine. In addition, the latency might affect the dosimetric accuracy of the plan delivery when the gating function is implemented.^{30,31} Also, the snout of the Hitachi Probeat system is fixed, and additional beam modifiers such as range shifters are inserted internally within the snout without changing the distance between the snout and the isocenter. Hence, the magnitude of the impact of a movable snout on the FOV is not reported in this study. Lastly, ideally the

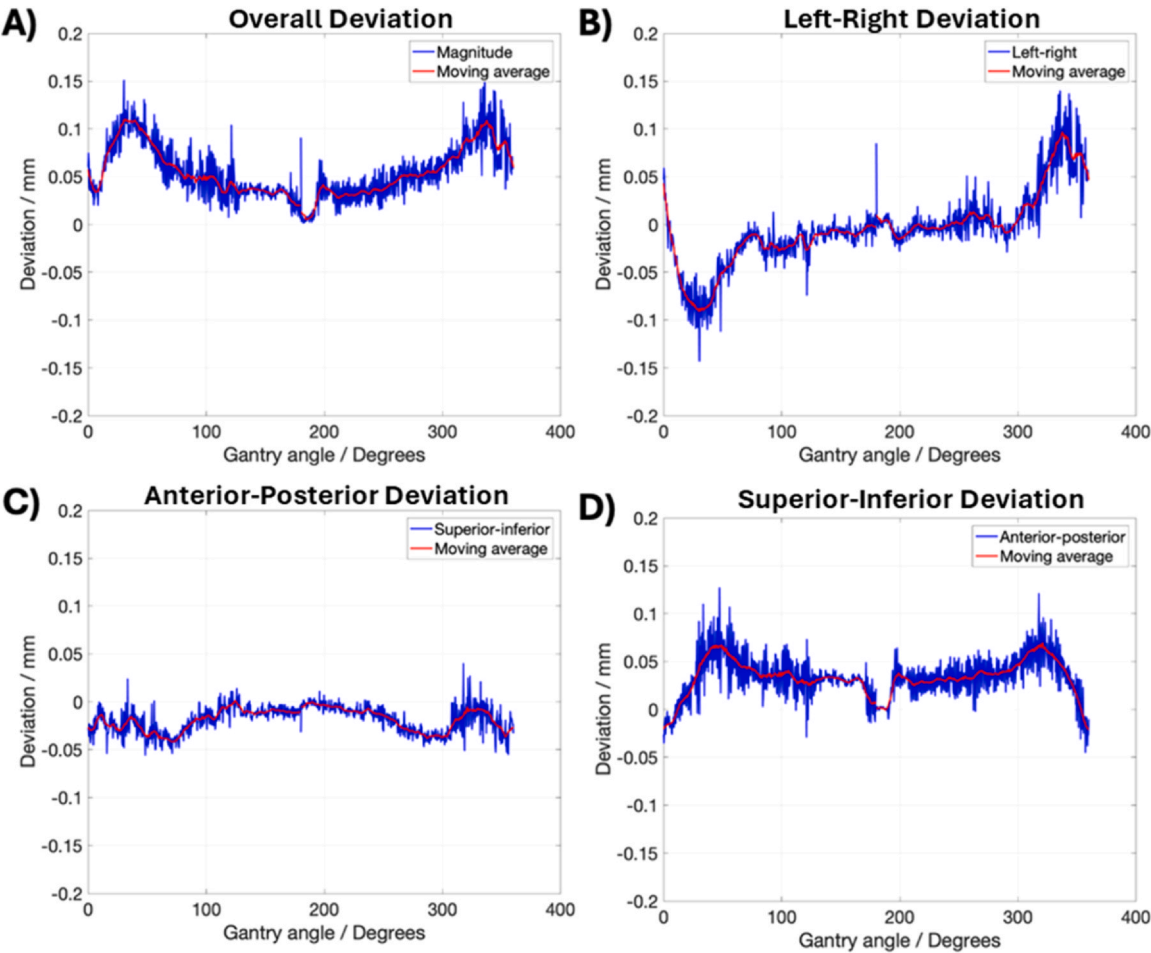


Figure 7. *Stability upon gantry rotation.* Show the translational deviation upon one full gantry rotation for the three cardinal directions and the overall magnitude. The raw data and moving averages (over 20 samples) are displayed by the blue and red lines, respectively. (A) Overall drift, (B) left-right drift, (C) anterior-posterior drift, and (D) superior-inferior drift.

Table 2
Table of tests and frequency for SGRT system.

Frequency	Test Category	Methods	Accuracy
Daily	SGRT and laser isocenter	Daily QA phantom positioned on couch at (0, 0, 0)	< 0.5 mm, 0.2°
Monthly	Static localization	Agreement with movement of couch: X, Y, & Z: -100 to 100 mm Pitch & Roll: -5° to 5°Yaw: -90° to 90°	< 1.0 mm, 0.1°
	Dynamic localization	Localization test by comparing with ANZAI laser system	< 1.0 mm difference between both system
Annual	Stability (Drift & Reproducibility)	Measurement over 200 min	< 2.0 mm
	Stability (Extensive)	Stability over entire gantry rotation: 179.9° to 180.0°	< 1.0 mm
		End-to-end test (Isocenter coincidence check)	< 1.0 mm

static localization test should be performed with a phantom capable of translating and rotating precisely independently of the couch. However, due to limited availability of a commercial phantom, this test was performed by moving the couch instead. Since the couch’s translation and rotational accuracy was checked as part of the weekly and monthly QA in accordance with TG-224, the decision to use the couch motion for SGRT commissioning was reasonable.

Conclusion

In this commissioning report, the LUNA 3D SGRT system has a relatively good machine tolerance, which is well within the recommendations of TG-147 and TG-302. We also reported on the different checks and frequencies that should be performed for the SGRT

system, which can serve as a reference for other Hitachi centers that are adding SGRT to their workflow.

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Author contributions

Study conception and design: Hong Qi Tan, Clifford Ghee Ann Chua, Kah Seng Lew, Data acquisition and analysis: Hong Qi Tan, Clifford Ghee Ann Chua, Kah Seng Lew, Wei Yang Calvin Koh, Data

interpretation: All authors, Statistical analyses: Hong Qi Tan, Kah Seng Lew, Obtained funding: Hong Qi Tan, Administrative, technical, or material support: Hong Qi Tan, Study supervision: Hong Qi Tan, Sung Yong Park, Drafting of manuscript: Hong Qi Tan, Kah Seng Lew, Clifford Ghee Ann Chua, James Kuan Huei Lee, Approval of final manuscript: All authors.

Declaration of Conflicts of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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